

SENNYBRIDGE, United Kingdom

WMO: 035070

Lat: 52.0633N

Lon: 3.6147W

Elev: 307

StdP: 97.69

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.7	-4.0	-7.4	2.1	-4.5	-5.8	2.4	-3.0	15.0	8.6	13.6	8.4	1.7	30	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	23.0	17.5	21.1	16.3	19.4	15.3	18.2	21.8	17.0	19.7	16.1	18.3	3.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.4	19.3	16.0	11.8	17.9	15.2	11.2	17.1	52.4	21.9	48.8	19.9	46.3	18.3	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.4	9.1	DB	-9.1	26.0	2.3	2.0	-10.7	27.5	-12.1	28.6	-13.3	29.7	-15.0	31.1	(4)
(5)			WB	-9.1	20.4	2.3	1.5	-10.8	21.5	-12.1	22.3	-13.4	23.1	-15.0	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	3.2	3.5	5.0	7.0	9.8	12.4	14.0	13.9	12.0	9.4	5.9	3.6	(6)	
	DBStd	4.89	3.34	3.15	2.83	2.84	2.78	2.49	2.25	2.25	2.30	2.88	3.16	3.73	(7)	
	HDD10.0	1068	211	182	157	96	37	6	1	1	7	46	127	199	(8)	
	HDD18.3	3655	469	416	414	339	264	178	136	139	190	278	374	457	(9)	
	CDD10.0	459	0	0	1	7	31	78	124	122	67	26	3	1	(10)	
	CDD18.3	4	0	0	0	0	0	1	2	2	0	0	0	0	(11)	
	CDH23.3	43	0	0	0	0	1	9	21	13	0	0	0	0	(12)	
	CDH26.7	3	0	0	0	0	0	0	2	1	0	0	0	0	(13)	
(14) Wind	WSAvg	3.5	4.1	4.1	3.7	3.4	3.4	3.1	2.8	2.8	3.0	3.7	3.6	3.8	(14)	
(15) Precipitation	PrecAvg	1539	173	136	104	93	101	87	94	106	104	171	173	189	(15)	
	PrecMax	1970	279	339	206	180	161	194	234	185	214	338	344	380	(16)	
	PrecMin	1115	17	40	34	22	39	35	29	27	24	64	90	63	(17)	
	PrecStd	227	70	74	44	44	34	44	51	36	50	73	65	83	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	11.8	15.0	19.7	22.3	24.7	25.8	25.2	21.8	17.3	13.2	11.9	(19)	
		MCWB	10.0	9.7	10.5	13.3	16.4	18.6	18.7	18.6	17.4	15.1	12.5	11.3	(20)	
	2%	DB	9.7	9.7	12.0	16.2	19.4	21.7	23.3	22.0	19.1	15.5	12.2	10.8	(21)	
		MCWB	9.3	8.9	9.2	11.4	14.5	16.4	17.5	17.4	15.7	14.2	11.6	10.5	(22)	
	5%	DB	8.9	8.8	10.2	13.8	17.2	19.5	20.9	19.8	17.4	14.5	11.3	9.8	(23)	
		MCWB	8.5	8.2	8.6	10.0	13.3	15.1	16.2	16.0	14.6	13.3	10.7	9.4	(24)	
	10%	DB	8.1	8.0	9.2	11.8	15.1	17.7	18.9	18.1	16.2	13.6	10.3	8.9	(25)	
		MCWB	7.6	7.4	8.0	9.2	11.8	14.2	15.3	15.3	14.1	12.6	9.7	8.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	10.2	11.0	13.9	17.0	19.3	19.4	19.7	18.4	15.8	12.8	11.5	(27)	
		MCDB	10.4	10.8	13.0	18.6	21.0	23.1	24.3	23.6	20.6	16.8	13.0	11.7	(28)	
	2%	WB	9.5	9.2	9.9	11.9	15.2	16.8	17.9	18.0	16.5	14.5	11.8	10.6	(29)	
		MCDB	9.7	9.6	11.2	15.3	18.7	20.4	21.9	21.1	18.1	15.3	12.1	10.8	(30)	
	5%	WB	8.6	8.4	9.1	10.6	13.8	15.5	16.8	16.8	15.4	13.6	10.8	9.6	(31)	
		MCDB	8.8	8.8	10.0	12.8	16.4	18.7	19.8	18.7	16.7	14.2	11.1	9.9	(32)	
	10%	WB	7.7	7.5	8.3	9.6	12.6	14.5	15.8	15.9	14.5	12.8	9.8	8.6	(33)	
		MCDB	8.0	8.0	9.1	11.5	14.6	17.0	18.1	17.4	15.7	13.4	10.3	8.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	5.5	6.7	8.2	8.5	8.4	8.2	8.0	7.8	6.2	5.4	4.9	(35)	
		MCDBR	5.0	5.9	8.8	11.9	11.8	12.5	12.6	10.8	10.0	6.5	5.2	4.9	(36)	
	5% WB	MCWBR	5.1	5.0	6.4	7.7	7.7	7.9	7.6	6.9	6.9	5.1	4.8	5.1	(37)	
		MCWBR	4.8	5.4	7.7	9.8	10.7	10.9	10.6	8.7	8.1	5.7	4.9	4.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.316	0.330	0.358	0.379	0.385	0.376	0.376	0.361	0.352	0.339	0.315	0.305	(40)	
		taud	2.328	2.371	2.297	2.261	2.280	2.330	2.335	2.394	2.416	2.426	2.388	2.303	(41)	
	Ebn at Noon		677	785	828	855	866	876	870	866	830	764	677	616	(42)	
		Edh at Noon	63	80	104	121	125	120	117	105	92	75	59	55	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.75	1.42	2.56	3.95	4.89	5.26	4.89	4.03	3.04	1.71	0.90	0.57	(44)	
	RadStd		0.07	0.16	0.30	0.43	0.28	0.55	0.50	0.32	0.23	0.15	0.09	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page